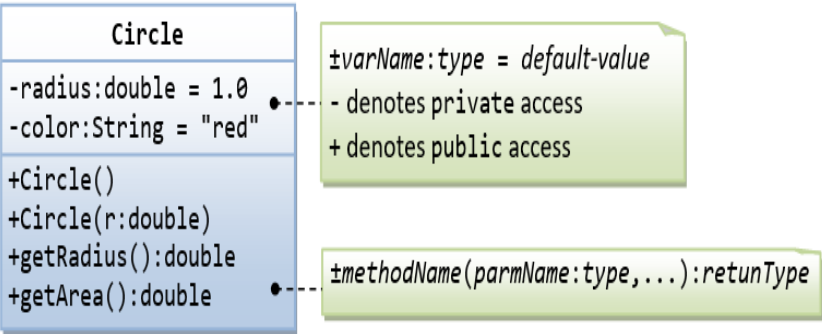


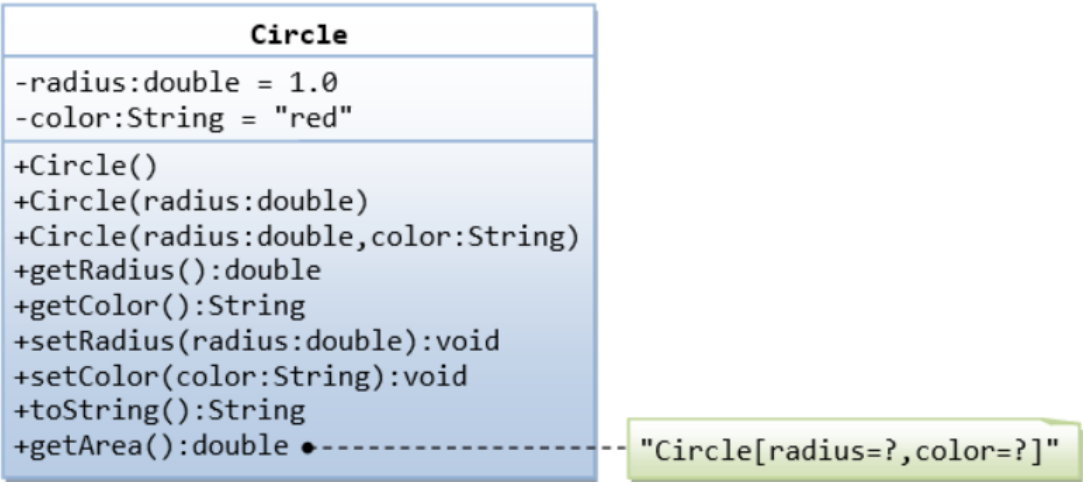
This first exercise shall lead you through all the *basic concepts* in OOP.



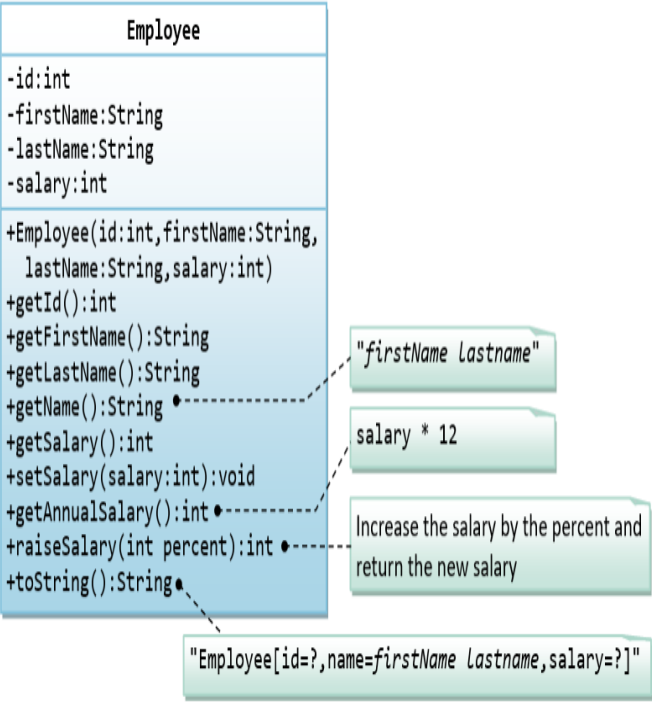
A class called **circle** is designed as shown in the following class diagram. It contains:

- Two private instance variables: radius (of the type double) and color (of the type String), with default value of 1.0 and "red", respectively.
- Two *overloaded* constructors - a *default* constructor with no argument, and a constructor which takes a double argument for radius.
- Two public methods: getRadius() and getArea(), which return the radius and area of this instance, respectively.

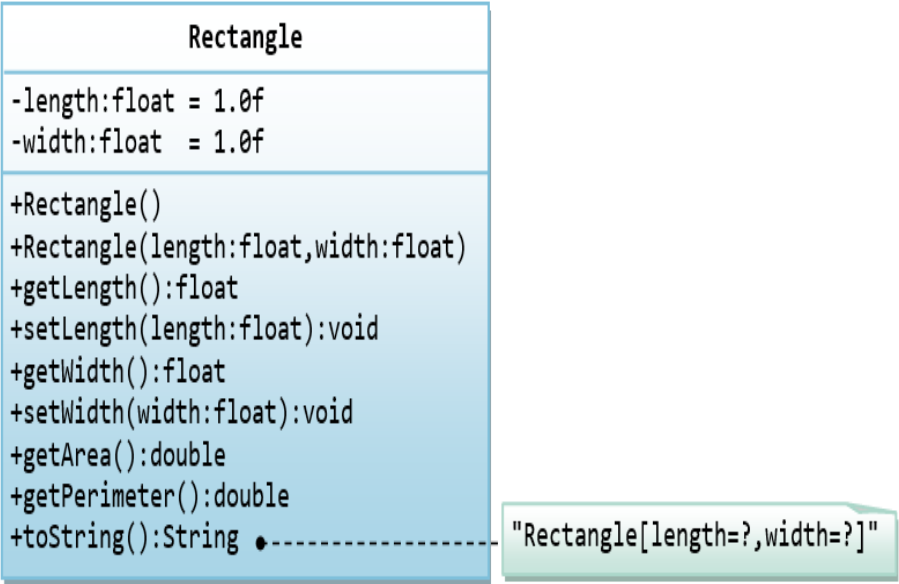
The final class diagram for the Circle class is as follows:



A class called Employee, which models an employee with an ID, name and salary, is designed as shown in the following class diagram. The method raisesSalary(percent) increases the salary by the given percentage. Write the Employee class.



A class called Rectangle, which models a rectangle with a length and a width (in float), is designed as shown in the following class diagram. Write the Rectangle class.



A class called InvoiceItem, which models an item of an invoice, with ID, description, quantity and unit price, is designed as shown in the following class diagram. Write the InvoiceItem class.

